How Options Markets Really Work

When you watch the stock market move up or down, you probably assume it is because investors changed their minds about the economy. Sometimes that is true. But a lot of the time, especially on a single day, the market is moving because of something much more mechanical. Something that has nothing to do with the economy at all.

The Two Players

There are two types of players that matter in the options market.

Dealers are big banks and market makers. Think Goldman Sachs or JPMorgan. Their job is to sell options to everyone who wants to buy them. They do not bet on market direction. They just want to earn a small fee on every trade. To do this safely, they constantly buy and sell the underlying index to neutralize their risk. This buying and selling creates real market moves.

Funds are the buyers, like hedge funds, pension funds and big asset managers. They buy options either to make money if the market moves in their favour, or to protect themselves if the market falls.

Here is the key insight: every time a fund buys an option, the dealer who sold it must trade the underlying index to protect themselves. That trading is often bigger than the original option trade. It moves markets.

What Is an Option?

An option is a contract. It gives you the right to buy or sell something at a fixed price in the future.

A **call option** is the right to buy. You buy a call when you think the market will go up.

A **put option** is the right to sell. You buy a put when you think the market will go down or when you want insurance against a crash.

The price you pay for an option is called the premium. Think of it exactly like an insurance premium. The riskier the situation, the more expensive the insurance.

Implied Volatility

Implied volatility is simply how expensive options are at any given moment. It reflects how much fear or uncertainty exists in the market.

When markets are calm, IV is low. Options are cheap. Nobody is rushing to buy insurance.

When markets are scared, IV is high. Options are expensive. Everyone wants protection.

You can think of IV like the price of a fire extinguisher. When there is no fire risk, extinguishers are cheap and nobody thinks about them. The moment someone smells smoke, the price skyrockets.

One important thing: options on downside moves are always more expensive than options on upside moves. This is because markets fall much faster than they rise. A crash can wipe out years of gains in a few weeks. A rally of the same size takes much longer. The market permanently prices this asymmetry — downside protection always costs more.

The Greeks

Dealers use numbers called Greeks to measure exactly how risky their positions are. Three of them matter for our story.

Delta — how much does the option move with the market?

If an option has a delta of 0.6, it moves €0.60 for every €1 the market moves.

For dealers, delta is also their hedge ratio. If they sold you a call with delta 0.6, they need to buy €0.60 worth of the index for every €1 of option they sold, otherwise they lose money when the market rises. This is called delta hedging. It happens automatically, constantly, all day long.

Gamma — how fast does delta change?

Delta is not a fixed number. It changes as the market moves. Gamma measures how fast it changes.

High gamma means the dealer's hedge needs constant adjustment. Every small market move forces them to buy or sell the index again. This is costly and creates real market flows.

The closer an option is to expiry and to the current market level, the higher its gamma. This is why options expiring today, zero-day options are so powerful.

Vanna — how does delta change when fear changes?

This one is the most subtle. Vanna measures how much delta changes when implied volatility moves, not when the market price moves, but when the fear level changes.

Here is an example. A dealer sold you a put option. Right now, that put has a delta of -0.3. Then suddenly fear spikes and so IV jumps. Now that same put has a delta of -0.5, even though the market has not moved yet. The dealer must immediately sell more of the index to rebalance their hedge. That selling pushes the market down, which creates more fear, which raises IV more, which forces more selling. This is the vanna spiral.

Net Gamma Exposure — GEX

GEX is a single number that tells you whether dealers are currently stabilizing the market or destabilizing it.

When GEX is **positive**, dealers are long gamma. Their hedging goes against the market move — they sell when the market rises and buy when it falls. This acts like a natural shock absorber. Markets stay calm and ranges stay tight.

When GEX is **negative**, dealers are short gamma. Their hedging goes with the market move — they buy when the market rises and sell when it falls. This amplifies every move. Small selloffs become large ones. Small rallies overshoot.

The **gamma flip level** is the market price where GEX crosses from positive to negative. Above this level, the market has a mechanical shock absorber. Below it, every move gets amplified. Institutional desks track this level every single day.

Open Interest

Open interest is simply the total number of option contracts that have not expired yet. It tells you where large positions are sitting in the market.

Strikes with very high open interest act like gravitational magnets. As expiration approaches, dealers are constantly adjusting their hedges around these strikes, which pulls the market toward them. This is called **gamma pinning**, the market gets stuck near the strike with the most open interest.

This is why experienced traders always check where the big open interest clusters are before a major expiry date. The market often ends up right there.

Zero-Day Options — 0DTE

A zero-day option expires the same day it is traded. These have become enormously important in modern markets.

Because they expire in hours, their gamma is at the absolute maximum. Every small market move creates huge rehedging flows from dealers. This is why some days feel explosively volatile while others feel impossibly calm, it often comes down to who is holding what 0DTE positions and on which side.

When funds sell 0DTE options to dealers, dealers become long gamma and the market gets calmer. When funds buy 0DTE options from dealers, dealers become short gamma and the market gets more volatile. The same instrument creates completely opposite effects depending on direction.

Vanna Flow

Vanna flow is the most counterintuitive concept here. It describes market moves caused purely by a change in fear, not by any change in actual prices.

When fear rises, IV spikes up

Put options gain sensitivity to market moves. Dealers who sold those puts must sell more of the index to stay hedged. That selling pushes the market down. More fear. More IV. More selling. The loop feeds itself. This is why selloffs often feel like they have a life of their own with no obvious news driving them.

When fear falls, IV collapses

Put options lose sensitivity. Dealers who were short the index to hedge their puts now have too much hedge. They must buy back. This creates a wave of mechanical buying with no fundamental reason behind it. The market can rally 2% on a day with literally no news — because dealers are unwinding hedges as volatility compresses.

This mechanical buying after a fear event is called the **vanna bid**. It is one of the most reliable short-term forces in equity markets.

The Full Mechanism

Now let us put it all together into one complete story. This is what a real institutional market event looks like beneath the surface.

The market opens calm. Dealers are long gamma. Every small move gets absorbed automatically. IV is low. Ranges are tight. Everything looks fine.

A large hedge fund starts buying put options aggressively. They are paying up and moving fast, a sign of urgency. Open interest at downside strikes starts building. Something is coming.

News breaks. Fear enters. IV starts rising. Because IV is rising, put options gain delta. Dealers must sell more of the index to stay hedged. The market starts drifting lower. Fear rises more. IV rises more. More selling. The vanna spiral begins.

The market falls below the gamma flip level. Now dealer hedging amplifies every move instead of absorbing it. A 0.5% drop becomes a 1.5% drop. The shock absorber is gone. The market is in freefall.

Near expiry, the market gravitates toward the strike with the most open interest. Dealers are constantly hedging around that level. The market gets pinned there.

The event resolves. Fear disappears. IV collapses, this is the vol crush. Dealers who were short the index to hedge their puts now have too much hedge. They buy back aggressively. The market rallies sharply. The news says it is relief. The reality is mechanical dealer buying triggered by falling volatility.